CoAtNet

Why local patterns and global context belong together

Paper analysis · Jake Van Slyke

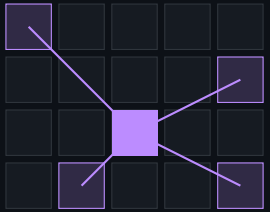
Dai, Liu, Le, and Tan · NeurIPS 2021

Two ways to gather visual information

Convolution

Reuse a local pattern detector
across the image

Attention



Choose relevant locations
using this image's features

The question: Can local structure help with limited data while attention preserves room to scale?

Schematic feature maps, not measured attention. Dai et al. (2021), §§1–2.1. Full attention can use every location.

Why they are natural complements

A reusable rule

The same learned weights
at every location and for every image.

Less to learn from each example

An adaptable rule

Weights change with the input
and can connect distant features.

More flexibility for complex relationships

Both mix features. They choose the weights differently.

CoAtNet's idea: Keep the spatial prior and the adaptable weighting.

Conceptual comparison of depthwise convolution and self-attention. Dai et al. (2021), Eqs. 1–3 and Table 1.

A range of designs between the endpoints

Convolution-only mixing — Attention-only mixing

Illustrative layouts: more attention stages toward the right

Designers can change **where** operations appear,
how far they reach, and **how** their weights depend on the input.

This is a **design space** with many possible hybrids.

Conceptual map based on Dai et al. (2021), §§2–3. C = convolution, T = attention; stems omitted. No numerical mixing scale.

CoAtNet makes two complementary choices

Inside an attention block

Content score + position bias

Learn how much each signal matters when mixing features.

Relative attention gives a spatial prior to adaptable weights.

Across the network

Choose where convolution gives way to attention.

Controlled comparisons guide the stage layout.

The contribution: a principled combination, tested across data scales.

Dai et al. (2021), §§2–4. Earlier work already used hybrids and relative attention; CoAtNet studies their combination.

One example: content plus relative position

i: location being updated

A is left, B is above,
C is farther right.

Candidate	A	B	C
Content match	2	1	0
Offset bias	-1	$+1$	0
Add the scores	1	2	0
Softmax share	24%	67%	9%

The bias is **learned for each relative offset**. These scores are invented.

Content favors A; adding position makes **B receive the largest share**.

Illustrative feature map and scores, not paper measurements. Softmax([1, 2, 0]) rounded; based on Dai et al. (2021), Eq. 3.

The equation follows the same recipe

$$\underbrace{A_{ij}}_{\text{mixing share}} = \mathrm{softmax}_j\left(\underbrace{q_i^\top k_j}_{\text{content match}} + \underbrace{b_{i-j}}_{\text{offset preference}}\right)$$

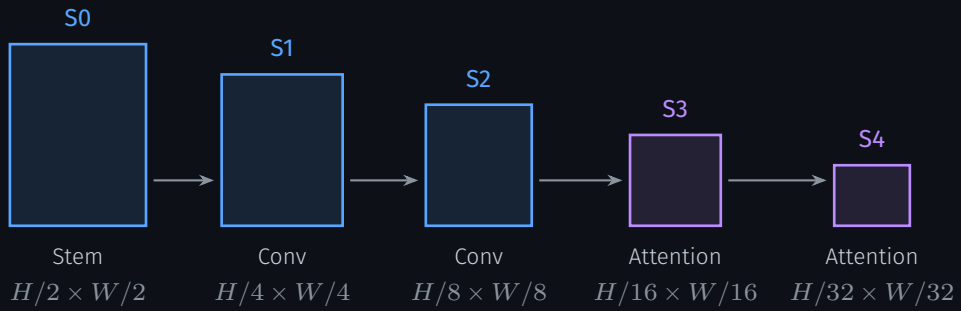
$$\underbrace{y_i}_{\text{updated feature}} = \sum_j \underbrace{A_{ij}}_{\text{share}} \underbrace{v_j}_{\text{feature to borrow}}$$

Our example: $y_i \approx 0.24v_A + 0.67v_B + 0.09v_C$

Score the candidates, turn scores into shares, then mix their features.

Adapted from Dai et al. (2021), Eq. 3. One attention head shown; scaling and projection details simplified.

Local processing before global mixing



Fewer locations make global attention cheaper. **C-C-T-T** names S1–S4.

Schematic stage sizes. Dai et al. (2021), §2.2 and Table 3. Conv stages use MBConv; attention includes relative bias.

What the layout experiments revealed

Plus a ViT-style stem
with relative attention

ImageNet-1K: about 1.28M images

Generalization: performance on unseen images.

More convolution generally helped.

JFT: over 300M images here

Capacity: ability to fit a large dataset.

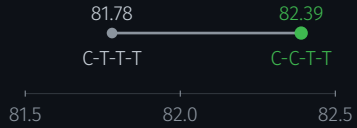
Two or three attention stages fit best.

C-C-T-T offered a useful balance.

Dai et al. (2021), §2.2 and Figure 1. Similar model sizes and compute; no augmentation or regularization in this study.

Two checks support the combination

Keep an extra convolution stage



+0.61 percentage points

JFT transfer, 224×224

Keep the relative-position bias

+0.4 percentage points

ImageNet-1K only, 384×384

Top-1 accuracy: first-choice label correct (%)

Zoomed axes; separate experiments

Redrawn from Dai et al. (2021), Tables 2 and 6. No repeated-seed uncertainty is provided in these tables.

The hybrid also scales beyond the ablations

ImageNet-1K only

86.0%

CoAtNet-3 matches NFNet-F5.

168M vs. 377M parameters

203B vs. 290B FLOPs

With larger pretraining datasets

88.56%

CoAtNet-4 roughly matches ViT-H/14's 88.55%.

12.7M vs. 300M pretraining images

The promise: useful spatial structure without giving up scaling potential.

Dai et al. (2021), Tables 4–5. ImageNet top-1. Models, resolutions, datasets, and recipes differ across comparisons.

How small is “small data”?

Evidence boundary: Strong results at large scale leave very small datasets untested.

Open question: Does that balance still help with only 1,000 examples?

Critical reading of Dai et al. (2021), §§4–5. Evaluated dataset sizes are shown; this is not a measured learning curve.

CoAtNet's answer

Build local features with convolution → Connect them globally with attention

Relative position keeps a spatial prior in the attention weights.

The experiments support this balance
across the image-classification regimes tested.

Thank you. **Questions?**

Dai et al. (2021), CoAtNet: Marrying Convolution and Attention for All Data Sizes. NeurIPS 34, 3965–3977.